CHEATSHEET · QUICK REFERENCE

# Lab 1 — Server Form Factors, Racking, and Power Planning

Inventory a running system with dmidecode/lshw and plan a balanced, cooled, power-budgeted 42U rack the way a data-centre technician does.

## Hardware inventory commands

Read chassis, CPU, memory, bus and expansion data from the running OS.

```
dmidecode -t system | grep -E "Manufacturer|Product|Serial"
```
Identify make / model / serial

```
dmidecode -t chassis | grep -E "Type|Height|Power"
```
Chassis type and U height

```
lscpu | head -20
```
CPU sockets, cores, model

```
lsmem || cat /proc/meminfo | head
```
Installed memory

```
lspci | head
```
Bus types (PCI / PCIe) and expansion cards

```
lshw -short 2>/dev/null | head -40
```
One-line hardware summary

## Rack layout rules (objective 1.1)

| Rule | Why |
|---|---|
| Heaviest equipment at the bottom | Rack balancing / stability |
| Redundant PDUs on separate circuits | Survive a single circuit / feed loss |
| Hot-aisle / cold-aisle alignment | Efficient cooling airflow |
| KVM 1U at top with cable arm | Crash-cart reachability, serviceability |
| Never load a circuit beyond 80% | Breaker headroom / inrush safety |

## Power budget sanity check

Sum nameplate watts and compare to PDU / breaker rating.

```
12 servers x 450 W + switch 250 W + KVM 30 W = 5,680 W
```
Total nameplate load

```
30 A @ 208 V = 6,240 W; 5,680 W = 91% -> OVER budget
```
Split load across 2 PDUs

## HCL check-before-buy workflow

| Step | Action |
|---|---|
| 1 | Get model string: dmidecode -s system-product-name |
| 2 | Open vendor HCL (Dell/HPE/Lenovo) or ubuntu.com/certified |
| 3 | Confirm CPU family, RAID controller, NIC are listed for the target OS |